

Homework 5

3.10 电路如图 P3.10 所示, $T_1 \sim T_5$ 的电流放大系数分别为 $\beta_1 \sim \beta_5$, b-c 间动态电阻分别为 $r_{be1} \sim r_{be5}$, 写出 A_u 、 R_i 和 R_o 的表达式。

3.10 The circuit is shown in Figure P3.10. The current amplification coefficients of $T_1 \sim T_5$ are $\beta_1 \sim \beta_5$, and the dynamic resistance between b-e is $r_{be1} \sim r_{be5}$. Write the expressions for A_u , R_i and R_o .

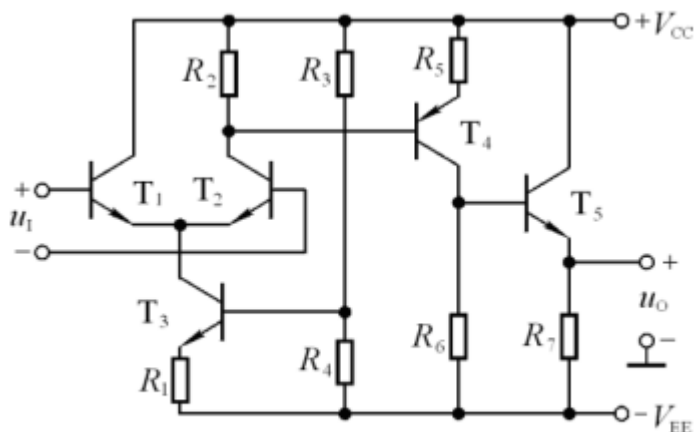


图 P3.10

3.11 电路如图 P3.11 所示。已知电压放大倍数为-100，输入电压 u_i 为正弦波， T_2 和 T_3 管的饱和压降 $U_{CES}=1V$ 。试问：

(1)在不失真的情况下, 输入电压最大有效值 U_{imax} 为多少伏?

(2) 若 $U_i = 10\text{mV}$ (有效值), 则 $U_o = ?$ 若此时 R_3 开路, 则 $U_o = ?$ 若 R_3 短路, 则 $U_o = ?$

3.11 The circuit is shown in Figure P3.11. It is known that the voltage amplification factor is -100, the input voltage u_I is a sine wave, and the saturation voltage drop of the T_2 and T_3 tubes is $U_{CES}=1V$. Ask:

(1) Without distortion, what is the maximum effective value U_{max} of the input voltage?

(2) If $U_i = 10\text{mV}$ (effective value), then $U_o = ?$ If R_3 is open at this time, $U_o = ?$ If R_3 is short-circuited, then $U_o = ?$

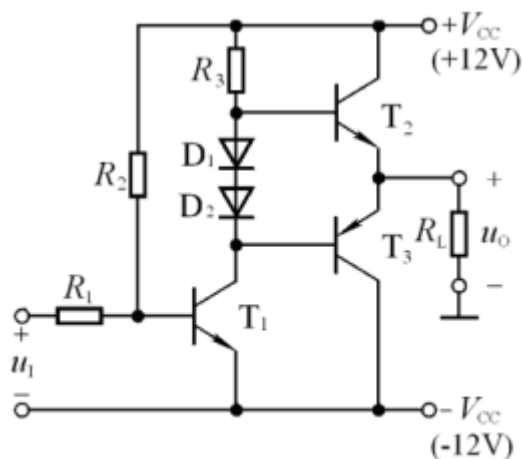


图 P3.11

4.3 多路电流源电路如图 P4.3 所示,已知所有晶体管的特性均相同, U_{BE} 均为 $0.7V$ 。试求 I_{C1} 、 I_{C2} 各为多少。

4.3 The multiple current source circuit is shown in Figure P4.3, assuming that the characteristics of all transistors are the same, and U_{BE} is $0.7V$. What is I_{C1} and I_{C2} ?

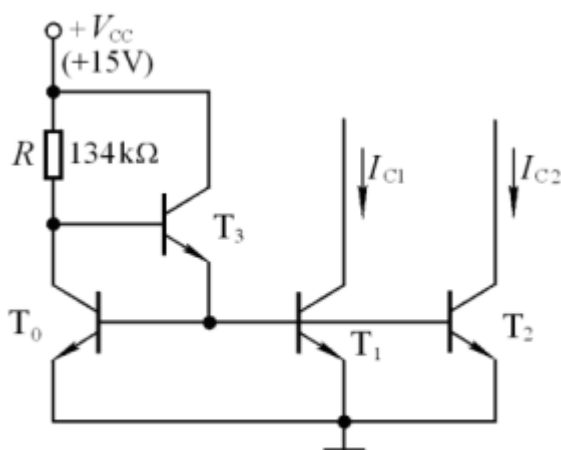


图 P4.3

4.4 电路如图 P4.4 所示, T_1 管的低频跨导为 g_m , T_1 和 T_2 管 d-s 间的动态电阻分别为 r_{ds1} 和 r_{ds2} 。试求解电压放大倍数 $A_u = \Delta u_o / \Delta u_i$ 的表达式。

4.4 The circuit is shown in Figure P4.4. The low-frequency transconductance of the T_1 tube is g_m , and the dynamic resistance between the T_1 and T_2 tubes d-s is r_{ds1} and r_{ds2} , respectively. Try to write the expression of voltage amplification factor $A_u = \Delta u_o / \Delta u_i$.

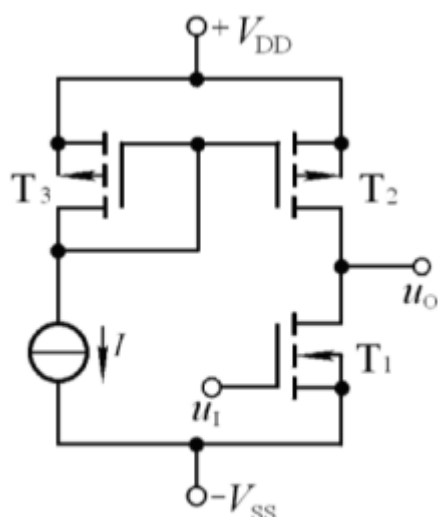


图 P4.4

4.7 电路如图 P4.7 所示， T_1 和 T_2 管的特性相同，所有晶体管的 β 均相同， R_{c1} 远大于二极管的正向电阻。当 $u_{i1} = u_{i2} = 0V$ 时， $u_o = 0V$ 。

(1) 求解电压放大倍数的表达式； (2) 当有共模输入电压时， $u_o = ?$ 简述理由。

4.7 The circuit is shown in Figure P4.7. The characteristics of T_1 and T_2 are the same, β of all transistors are the same, and R_{c1} is much larger than the forward resistance of the diode. When $u_{i1} = u_{i2} = 0V$, $u_o = 0V$.

(1) Solve the expression of the voltage amplification factor; (2) When there is a u_{ic} , $u_o = ?$ Explain the reason.

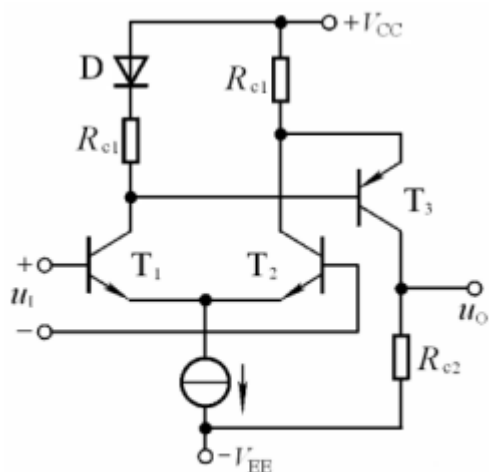


图 P4.7

4.10 比较图 P4.10 所示两个电路，分别说明它们是如何消除交越失真和如何实现过流保护的。

4.10 Compare the two circuits shown in Figure P4.10 and explain how they eliminate

crossover distortion and how to achieve overcurrent protection.

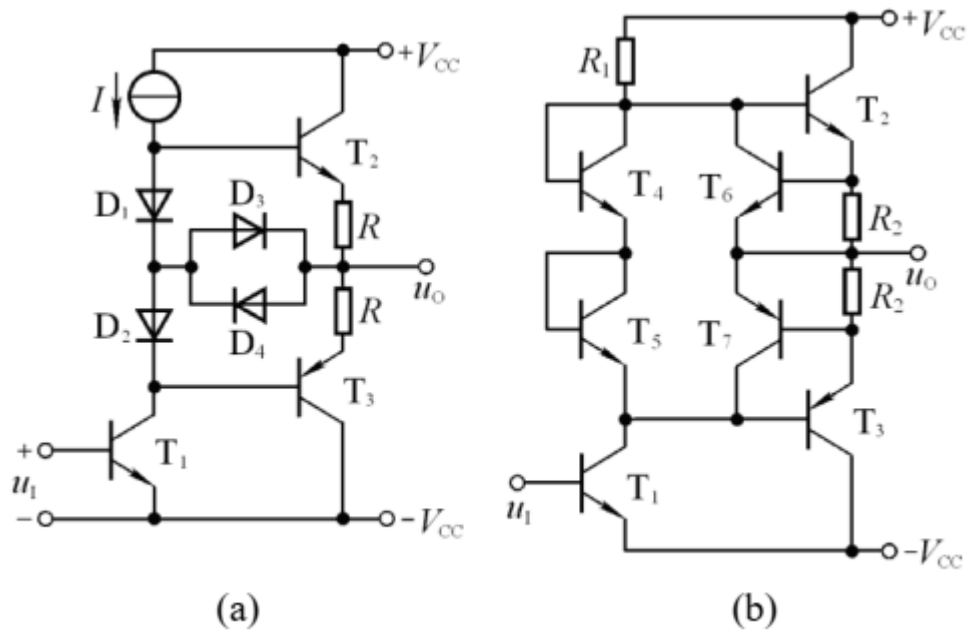


图 P4.10